Cross-Border IPS AI Agent

Federated FHIR terminology alignment for cross-border Type 2 diabetes summaries.

Judge-facing wording

FHIR IPS is the interoperable artifact; PDFs are human-readable renderings.

Aim

Make a local diabetes report usable across borders by converting it into a machine-readable, universally coded, HL7 FHIR IPS-style patient summary. FHIR IPS is the interoperable artifact; PDFs are human-readable renderings.

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Problem Statement

Local clinical language blocks interoperability. T2DM, sugar disease, madhumeha type 2, type 2 DM, and DM2 may refer to the same concept, but receiver systems need standard codes and predictable FHIR resources.

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Solution Flow

Doctor report -> clinical fact extraction -> local registry lookup -> federated terminology linker -> FHIR ConceptMap validation -> FHIR IPS document Bundle -> target-country report and readiness gaps.

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Machine-Readable Proof

The final exchange artifact is an IPS-style FHIR R4 document Bundle with Bundle.type=document, Composition first, coded resources, and representative validator evidence showing 0 errors.

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Universal Coding Proof

Local clinical terms are normalized into accepted healthcare codes: SNOMED CT and ICD-10 for conditions, LOINC for labs, RxNorm for medications, and FHIR resources for exchange structure.

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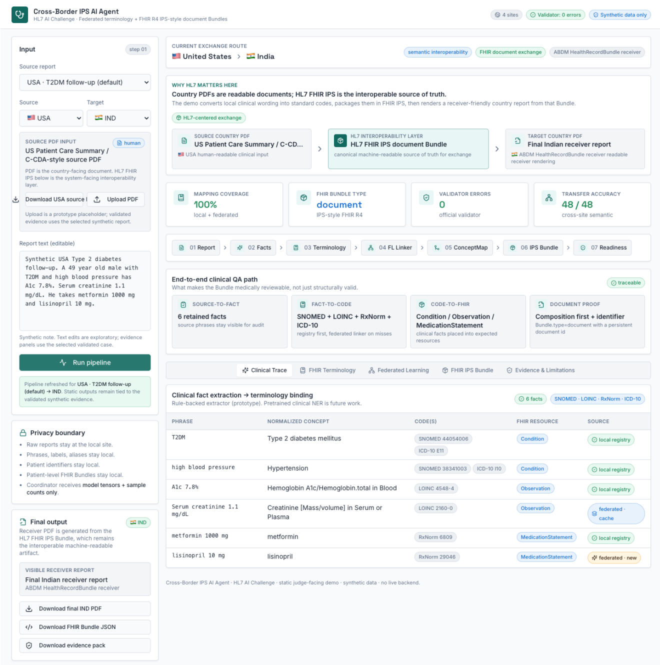
Each country/site trains a terminology linker locally. The coordinator receives model tensors and sample counts only; it does not receive raw reports, identifiers, labels, aliases, or patient-level FHIR Bundles.



Dashboard screenshot from automated click-test.

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FHIR-Native Global Linker

The model predicts candidate mappings, but HL7 makes them auditable through local CodeSystem, target ValueSet, ConceptMap, simulated translate, lookup, and validate-code.

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Cross-Border Sharing

The same FHIR IPS can support USA, India, Australia, and Europe routes. Judges can download source PDF, target PDF, FHIR Bundle JSON, and evidence pack JSON. Readiness checks are not certification.

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Evidence

20 synthetic reports, 4 country sites, 768 terminology training mentions, 192 globally unseen examples, 48/48 cross-site transfer examples correct, and 0 validator errors for representative IPS-style Bundles.

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Honest Limits

Synthetic data only. Rule-backed extraction. Simulated terminology operations. Readiness checks only. FedAvg gives data locality only and is not cryptography or formal de-identification.

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Final Claim

Cross-Border IPS AI Agent combines semantic interoperability and data interoperability. Federated learning aligns local terminology, FHIR terminology artifacts make mappings auditable, and FHIR IPS packages the result for cross-border exchange.

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